

1. [10 points] Explain the unnecessary redundancies of data stored in these csv files.

Ans. The data redundancies observed in the CSV files include:

- Instructor and Advisor Redundancy:** Since instructors are also advisors, there is redundant data for InstructorID and AdvisorID. This leads to duplication of data for names and IDs of individuals who serve both roles.
- Course Redundancy:** The same course can be taught by multiple instructors, leading to potential redundancy of course details like CourseName and CreditHours across multiple rows if a separate record is kept for each instructor teaching the course.
- Student Information Duplication:** If a student is enrolled in multiple courses, their details (like FirstName, LastName, Email, etc.) might be repeated in each row of the enrollments.csv file, leading to redundancy.

2. [20 points] Create an ER model to capture the information in these csv files. Draw the model, making sure to indicate key attributes, attributes (if any) and multiplicity of relationships, etc. The model should remove the data redundancies observed in the previous question.

